

② Matrices

- Characteristic equation: $\det(A - \lambda I) = 0$.
- Eigen values: solve the equation (characteristic equation).
- Eigen vectors: substitute back λ into matrix & calculate cross product.

• Diagonalization:

$A = XDX^{-1}$ $\rightarrow$ diagonal matrix, λ
 $\hookrightarrow$ eigenvectors

$X^{-1}AX = D \rightarrow$ Identity $\star$
 X is said to diagonalize A .

$$D = \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \lambda_n \end{bmatrix}$$

$$X^{-1}AX = D$$

$$XX^{-1}AX = XD$$

$$A = XDX^{-1}$$

$$\begin{aligned} Ae &= \lambda e \\ A^2e &= \lambda Ae \\ &= \lambda(\lambda e) \\ &= \lambda^2 e \end{aligned}$$

Prove using induction.

$$\begin{aligned} Ae &= \lambda e \\ AA^{-1}e &= \lambda A^{-1}e \\ e &= \lambda A^{-1}e \\ \therefore \frac{1}{\lambda} e &= A^{-1}e \end{aligned}$$

$$\therefore A^{-1}e = \frac{1}{\lambda} e$$

(Proofs)

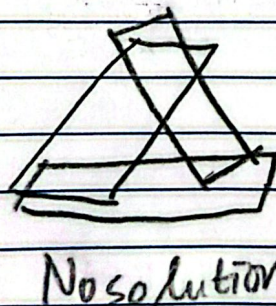
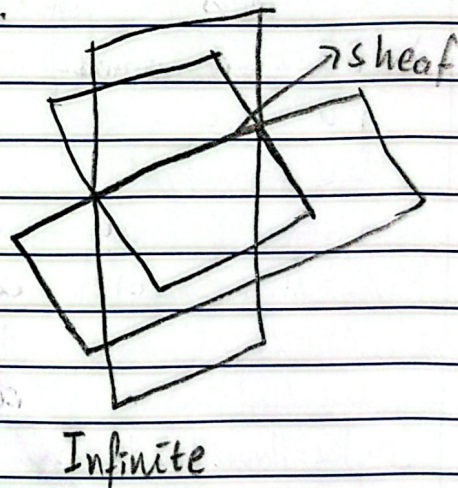
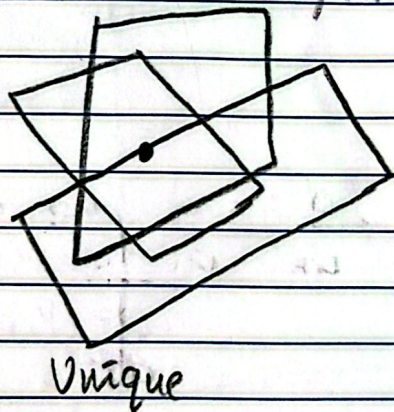
- Cayley-Hamilton Theorem: a square matrix always satisfies its own characteristic equation:

$$\begin{aligned} -\lambda^3 + 5\lambda^2 + 6\lambda - 8 &= 0 \\ \therefore -A^3 + 5A^2 + 6A - 8I &= 0 \end{aligned}$$

- If $P^{-1}AP = X$, use matrix multiplication to find A .

★ Singularity of solutions are important.

- Unique solution: $\det(M) \neq 0$. $\rightarrow$ has an inverse, non singular
Other cases: $\det(M) = 0$. $\rightarrow$ no inverse, singular
- Consistent v/s inconsistent system:
A consistent solution has a solution that matches with the three equations.
An inconsistent solution would lead to a contradiction.
- Unique $\rightarrow$ meet uniquely at a single point.
Inf $\rightarrow$ meet along a line/sheaf
No $\rightarrow$ Planes never meet at a point. Form triangular prism
- Always check for parallel planes using scale factors. Also through identical/non-identical planes.



(triangular prism)